Roll No.....

DRONACHARYA COLLEGE OF ENGINEERING, GURGAON

First Sessional Examination March 2025

(Gurugram University)

Semester- IV

Branch: CSE(AI&ML)

QUESTION PAPER

Subject with Code: Computer Organization and Architecture

Time:- 2 hours

Max Marks:50

Note: All questions are compulsory, each question carries 10 marks.

Q. 1. Objective type / Short-answer type questions. ***(2.5x4=10)***

(i) Conversions: [CO1]

a) 1110101 (Binary to Decimal) b) 7562 (Decimal to Octal)

(ii) Explain Three-State Buffers with graphical symbol. [CO1]

(iii) Define 16 Bits of instruction format. [CO1]

(iv) Explain following operations with example: [CO1]

a) Mask b) Selective Clear

Q 2. a) Represent the following: [CO2]

If (P=1) then (R1← R2) else if (Q=1) then (R1← R3)

b) Draw Arithmetic Logic Shift Unit with functional table. [CO1]

Q 3. Explain Floating Point Representation with Single Precision format with an example. [CO1]

Q4. a) List and explain the basic registers used in a computer along with their sizes. [CO2]

b) Draw Common Bus System for computer architecture [CO2]

Q5. Explain **timing signals** in the execution of instructions in a computer system. Draw a **flowchart** illustrating the sequence of timing signals for different types of instructions [CO2]

OR

What is an **interrupt** in a computer system? Draw a **flowchart** illustrating the **interrupt cycle**, detailing the sequence of events that occur when an interrupt is triggered. [CO2]